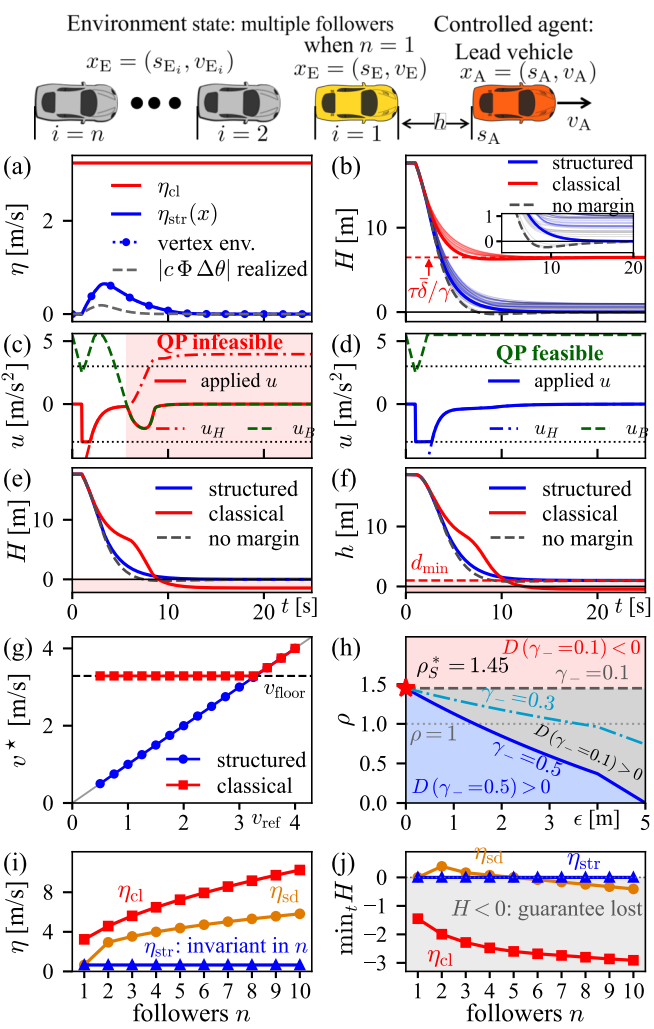

Environment state: multiple followers
$x_{\mathrm{E}} = (s_{\mathrm{E}_i}, v_{\mathrm{E}_i})$
$i = n$
$i = 2$
when $n = 1$
$x_{\mathrm{E}} = (s_{\mathrm{E}}, v_{\mathrm{E}})$
$i = 1$
Controlled agent:
Lead vehicle
$x_{\mathrm{A}} = (s_{\mathrm{A}}, v_{\mathrm{A}})$
h
s_{A}
v_{A}
(a)
η [m/s]
η_{cl}
$\eta_{\mathrm{str}}(x)$
vertex env.
$|c\,\Phi\,\Delta\theta|$ realized
(b)
H [m]
structured
classical
no margin
$\tau\bar{\delta}/\gamma$
(c)
u [m/s^2]
QP infeasible
applied u
u_H
u_B
(d)
QP feasible
(e)
(f)
h [m]
$d_{\min}$
t [s]
(g)
$v^\star$ [m/s]
v_{floor}
v_{ref}
(h)
ρ
$\rho_s^* = 1.45$
$D(\gamma_- = 0.1) < 0$
$\gamma_- = 0.1$
$\gamma_- = 0.3$
$\gamma_- = 0.5$
$\rho = 1$
$D(\gamma_- = 0.1) > 0$
$D(\gamma_- = 0.5) > 0$
ϵ [m]
(i)
η_{sd}
η_{str}: invariant in n
followers n
(j)
$\min_t H$
$H < 0$: guarantee lost